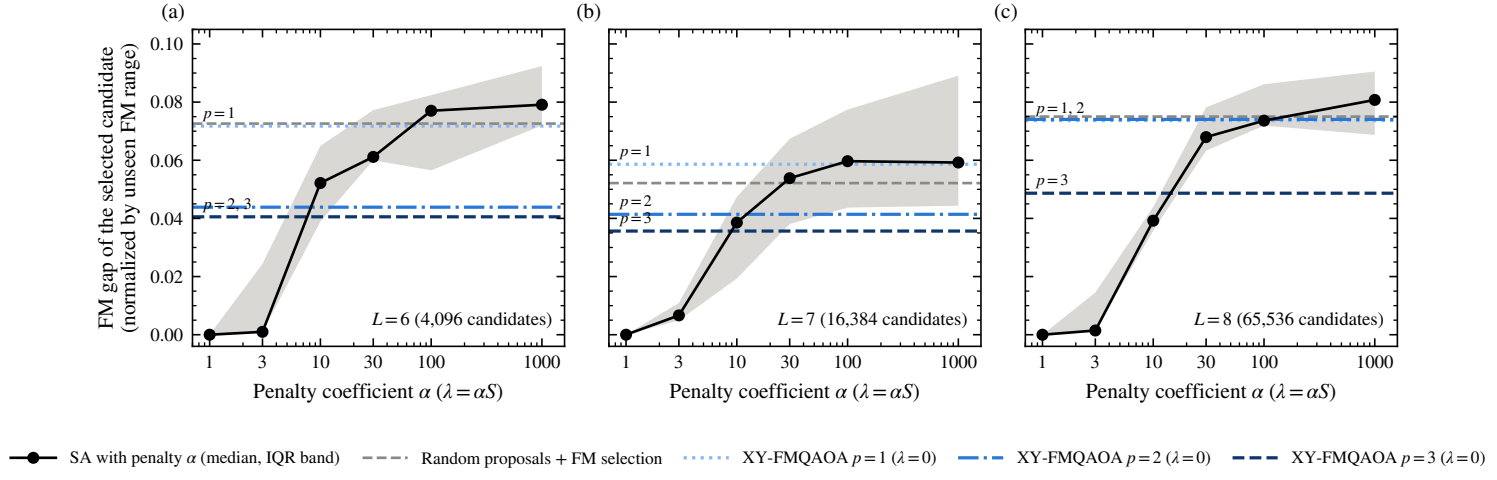
(a)
(b)
(c)
FM gap of the selected candidate
(normalized by unseen FM range)
0.10
0.08
0.06
0.04
0.02
0.00
p = 1
p = 2, 3
p = 1, 2
p = 3
L = 6 (4,096 candidates)
L = 7 (16,384 candidates)
L = 8 (65,536 candidates)
1
3
10
30
100
1000
Penalty coefficient α (λ = αS)
SA with penalty α (median, IQR band)
Random proposals + FM selection
XY-FMQAOA p = 1 (λ = 0)
XY-FMQAOA p = 2 (λ = 0)
XY-FMQAOA p = 3 (λ = 0)